Simulating Quantum Key Distribution Using the BB84 Protocol with Qiskit by Team Plancks-Konstant

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Part 1: Project Plan and Team Details

Team details

Name	Role
Promise Adesemoye	Research, Code implementation and testing
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Project aim

To simulate the BB84 quantum key distribution protocol in Qiskit, to show that Alice and Bob can establish a shared secret key, and to show that an intercept-resend eavesdropper (Eve) can be detected through an elevated error rate.

Objectives

1. Understand and document the BB84 protocol and the quantum principles it relies on.
2. Implement the full protocol in Qiskit: qubit preparation, transmission, measurement, sifting and error checking.
3. Implement an intercept-resend eavesdropper and measure the error rate she causes.
4. Compare the results against the theoretical error rate of 25%.

Deliverables

The Python script `bb84_final.ipynb`, its console output (circuit diagram, per-qubit tables, summary statistics) and this report.

Part 2: Project Presentation, Methodology, Results and Discussion

1. Introduction

Classical key exchange relies on computational assumptions, and an eavesdropper can copy classical data without leaving a trace. Quantum key distribution (QKD) instead relies on physics. BB84, proposed by Bennett and Brassard in 1984, lets two parties, Alice and Bob, agree on a secret key over an open quantum channel and detect any interception, because measuring a quantum state disturbs it.

2. Background

Three quantum facts underpin the protocol.

1. **Bases.** A qubit can be prepared and measured in different bases. BB84 uses the Z (rectilinear) basis with states |0> and |1>, and the X (diagonal) basis with states |+> and |->.
2. **Measurement.** Measuring in the same basis as the preparation gives the encoded bit with certainty. Measuring in the other basis gives a uniformly random result.
3. **Disturbance and no-cloning.** Measurement collapses the state, and an unknown quantum state cannot be copied perfectly. An eavesdropper therefore cannot both learn the bit and leave the qubit unchanged.

The encoding used is:

Bit	Z basis	X basis
0	|0>	|+>
1	|1>	|->

3. The BB84 protocol

1. **Preparation.** Alice generates random bits and, for each, a random basis, and prepares the matching qubit.
2. **Transmission.** The qubits are sent to Bob.
3. **Measurement.** Bob picks a random basis for each qubit and measures.
4. **Sifting.** Alice and Bob publicly announce their bases (not their bits) and keep only the positions where the bases agree, about 50% of the qubits.
5. **Error check.** They publicly compare a random subset of the sifted bits. A high error rate indicates interference and the key is discarded.
6. **Key.** The bits not revealed in the comparison form the shared secret key.

Why interception is detectable. In an intercept-resend attack Eve measures each qubit in a randomly chosen basis and resends a new qubit. She guesses the basis correctly half the time, which causes no error. When she guesses wrong (the other half), the resent qubit is in the wrong basis, and if Bob's basis matches Alice's his result is wrong with probability 1/2. The expected error rate on the sifted key is therefore 0.5 x 0.5 = **25%**.

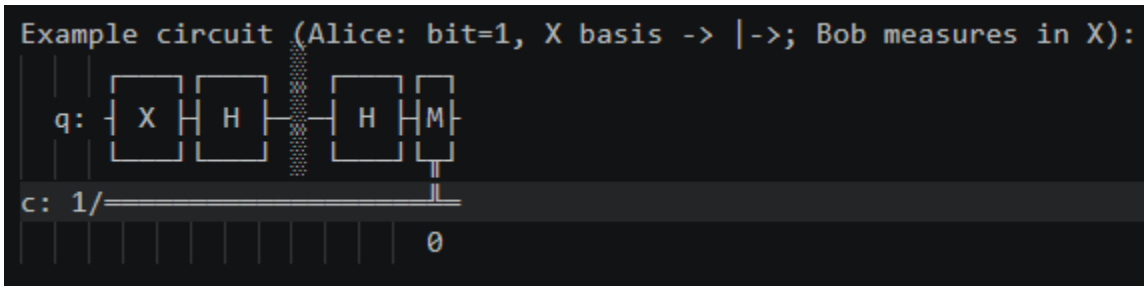
4. Methodology

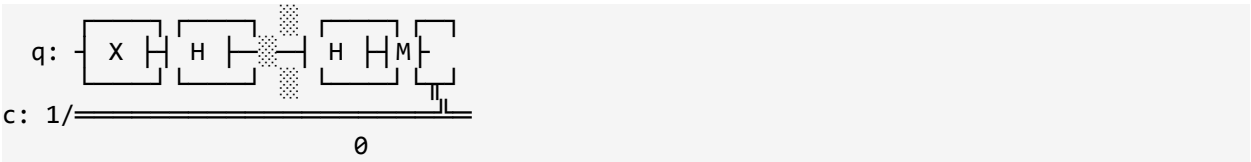
Tools. Python, Qiskit 2.6.0 and the Qiskit Aer simulator (`AerSimulator`). Each qubit is simulated as an independent one-qubit circuit run for a single shot, because in the real protocol each qubit is sent and measured exactly once.

Circuit construction.

- *Encoding:* if the bit is 1, apply an `X` gate; if the basis is X, then apply an `H` gate.
- *Measurement in Z:* measure directly.
- *Measurement in X:* apply `H`, then measure.
- *Eve:* uses the same measurement step with her own random basis, then re-encodes the bit she obtained in that basis and passes the new qubit on.

The circuit for a qubit encoded as |-> (bit 1, X basis) and measured by Bob in the X basis is:





The barrier (⋮) marks the quantum channel, which is where Eve would intervene.

Procedure. The function `bb84()` runs the six protocol steps for a chosen number of qubits, with Eve switched on or off. In the error check, 50% of the sifted bits are compared and then discarded. The decision threshold is set at 11% error, a commonly cited bound that allows for some real-world noise. For each qubit the program records Alice's bit, basis and state, Eve's basis and result, Bob's basis and result, and whether the position was kept and matched.

Experiments.

1. A 400-qubit run without Eve (fixed seed).
2. A 400-qubit run with Eve (same seed).
3. Twenty repeated 200-qubit runs with Eve, to average the error rate.

5. Results

Table 1: Summary of the 400-qubit runs

Measure	Without Eve	With Eve
Qubits sent	400	400

Measure	Without Eve	With Eve
Kept after sifting	182	201
Bits compared publicly	91	100
Errors found	0	26
Error rate	0.0%	26.0%
Verdict	Channel clean, key kept	Eavesdropper detected, key discarded
Final keys identical	Yes	No

Table 2: Error rate with Eve over repeated runs

Quantity	Value
Runs	20 (200 qubits each)
Average error rate	25.7%
Theoretical value	25%

Sifting rate. 182 of 400 (45.5%) and 201 of 400 (50.3%) positions survived sifting, consistent with the expected 50%.

Per-qubit behaviour. The tables printed by the program show the mechanism directly. In the run without Eve, every kept qubit matches (for example qubits 1, 3, 8, 9, 11, 14 and 19 of the first 20). In the run with Eve, errors appear only where Eve guessed the wrong basis. For example, qubit 0: Alice sent |0> in the Z basis, Eve measured in X and read a random 1, and resent a state that made Bob, who also used Z, read 1 instead of 0. Qubits 4, 7 and 16 follow the same pattern. Where Eve guessed correctly (for example qubits 6, 8, 9 and 11), no error appears.

```
#################### WITHOUT EVE ####################

Per-qubit outputs (first 20 of 400):
  #  | Alice bit basis state | Bob basis Bob bit | Keep? Match?
-------------------------------------------------------------
  0  |         0     Z   |0> |         X       0 |    no      -
  1  |         0     Z   |0> |         Z       0 |   yes    yes
  2  |         1     X   |-> |         Z       0 |    no      -
  3  |         0     X   |+> |         X       0 |   yes    yes
  4  |         1     X   |-> |         Z       0 |    no      -
  5  |         1     X   |-> |         Z       1 |    no      -
  6  |         1     Z   |1> |         X       0 |    no      -
  7  |         1     X   |-> |         Z       0 |    no      -
  8  |         0     Z   |0> |         Z       0 |   yes    yes
  9  |         0     Z   |0> |         Z       0 |   yes    yes
 10  |         1     X   |-> |         Z       1 |    no      -
 11  |         0     Z   |0> |         Z       0 |   yes    yes
 12  |         1     Z   |1> |         X       1 |    no      -
 13  |         1     X   |-> |         Z       1 |    no      -
 14  |         0     X   |+> |         X       0 |   yes    yes
 15  |         1     Z   |1> |         X       0 |    no      -
 16  |         1     Z   |1> |         X       1 |    no      -
 17  |         0     X   |+> |         Z       0 |    no      -
 18  |         0     X   |+> |         Z       0 |    no      -
 19  |         1     X   |-> |         X       1 |   yes    yes

=== Without Eve ===
Qubits sent:              400
After sifting:            182  (~50% expected)
Bits compared (public): 91
Errors found:             0
Error rate:               0.0%
Verdict:                  Channel looks clean -> keep key
Final key length:         91
Final keys identical:     True
Alice key (first 32):     00011101110111001101111000110110
Bob key   (first 32):     00011101110111001101111000110110
```

```
################### WITH EVE ###################

Per-qubit outputs (first 20 of 400):
  #  | Alice bit basis state | Eve basis Eve bit | Bob basis Bob bit | Keep? Match?
-------------------------------------------------------------------------------------
  0  |         0     Z   |0> |         X       0 |         Z       1 |   yes     NO
  1  |         0     Z   |0> |         Z       0 |         X       0 |    no      -
  2  |         1     X   |-> |         Z       0 |         Z       0 |    no      -
  3  |         0     X   |+> |         X       0 |         Z       0 |    no      -
  4  |         1     X   |-> |         Z       0 |         X       1 |   yes    yes
  5  |         1     X   |-> |         Z       1 |         Z       1 |    no      -
  6  |         1     Z   |1> |         X       0 |         Z       0 |   yes     NO
  7  |         1     X   |-> |         Z       0 |         X       0 |   yes     NO
  8  |         0     Z   |0> |         Z       0 |         Z       0 |   yes    yes
  9  |         0     Z   |0> |         Z       0 |         Z       0 |   yes    yes
 10  |         1     X   |-> |         Z       1 |         Z       1 |    no      -
 11  |         0     Z   |0> |         Z       0 |         Z       0 |   yes    yes
 12  |         1     Z   |1> |         X       1 |         X       1 |    no      -
 13  |         1     X   |-> |         Z       0 |         Z       0 |    no      -
 14  |         0     X   |+> |         X       0 |         Z       0 |    no      -
 15  |         1     Z   |1> |         X       1 |         X       1 |    no      -
 16  |         1     Z   |1> |         X       0 |         Z       0 |   yes     NO
 17  |         0     X   |+> |         Z       1 |         Z       1 |    no      -
 18  |         0     X   |+> |         Z       0 |         Z       0 |    no      -
 19  |         1     X   |-> |         X       1 |         Z       1 |    no      -

=== With Eve (intercept-resend) ===
Qubits sent:              400
After sifting:            201  (~50% expected)
Bits compared (public): 100
Errors found:             27
Error rate:               27.0%
Verdict:                  EAVESDROPPER DETECTED -> discard key
Final key length:         101
Final keys identical:     False
Alice key (first 32):     11011101010111011110000010111010
Bob key   (first 32):     10011110111111111111011001010101

Average error rate with Eve over 20 runs: 23.5%  (theory: 25%)
```

6. Discussion

Findings. The simulation reproduces the main property of BB84. Without interference the sifted keys agree perfectly, and with an intercept-resend eavesdropper the error rate rises to about 25%, matching theory (25.7% averaged over 20 runs; 26.0% in the single seeded run). The error rate in a single run varies because the compared sample is finite (100 bits), which is why the averaged figure is closer to 25%.

Interpretation. Eve gains information only on the qubits where she guesses the basis correctly, but she cannot tell which those are. Each wrong guess has a 50% chance of showing up as an error in the sifted key, so she cannot escape detection over a sufficiently large sample.

Limitations.

- The simulation is noiseless. Real channels and detectors introduce errors even without an eavesdropper, so the no-Eve error rate would not be exactly 0% and the threshold (here 11%) matters.
- Only the intercept-resend attack is modelled. Other attacks, such as photon-number-splitting attacks on imperfect single-photon sources, are not covered.
- Public communication is assumed to be authenticated. Without authentication, a man-in-the-middle attack would be possible.

- Error correction and privacy amplification, which a full QKD system needs to remove residual errors and any partial information held by Eve, are not implemented.

Possible extensions. Adding a noise model in Aer, varying the fraction of qubits Eve intercepts, plotting detection probability against sample size, and implementing a simple error-correction step.

7. Conclusion

A complete BB84 simulation was implemented in Qiskit. The results confirm that matching-basis measurements give identical keys, that sifting retains about half the qubits, and that intercept-resend eavesdropping produces roughly 25% errors, which makes the eavesdropper detectable.

Part 3: Declaration of Assistance and Resources

AI tools

Tool	How it was used
Claude (Anthropic, Claude Sonnet 5.5)	Explained the BB84 protocol and the underlying quantum concepts; wrote the first version of the Python/Qiskit simulation code which was later reviewed, corrected and re-edited by the team
ChatGPT	Research, further exploration of the topic

References and external materials

Source	How it was used
Bennett, C. H. and Brassard, G. (1984). Quantum cryptography: Public key distribution and coin tossing. *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, pp. 175-179.	Original definition of the protocol
Qiskit documentation (https://qiskit.org/documentation/)	Circuit construction and simulator usage

Software and libraries

Python 3, Qiskit 2.6.0 and Qiskit Aer (open source), used under their respective licences.

Mentorship and other assistance

Person	Role	Assistance
Chiagoziem Mushinwa	Mentor	Provided guidance and clarity on BB84 protocols